

Monty Hall and Independence

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MATH 241

Lecture #7 — Ch. 2.5-2.7

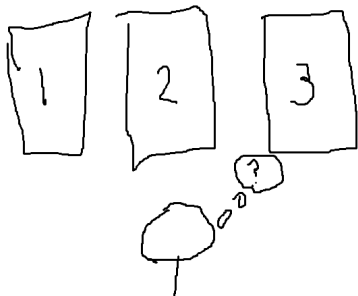
Reminders!

- Homework #3 is due on **Friday, 9/25** at 11:59pm
- Office hours this week:
 - ▶ Monday from 1-2:30pm
 - ▶ Tuesday from 3-4:30pm
 - ▶ Friday from 10:30-11:30am
- Exam #1 will be held **next Tuesday, 9/29** during lecture
 - ▶ More about this at the end of lecture

Today:

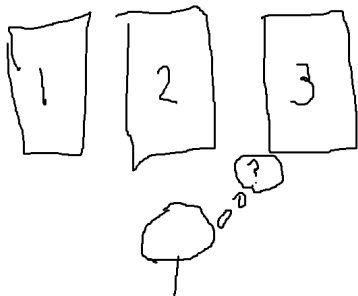
- Can we improve our chances at picking a prize from a random door?
 - ▶ Monty Hall problem: **investigate** the problem and how we can use conditional probability concepts to inform our decision-making
- What is independence? (from a statistical perspective)
 - ▶ **Define** independence statistically, and identify how previously-learned properties change as a result

Monty Hall Problem



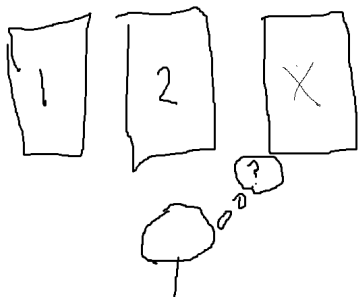
- Let's say you're on Let's Make a Deal and you have to pick a door. Behind one of them is a brand new car, while junk is behind the other doors.

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Monty Hall Problem



- Before revealing if you're right, Wayne (the host) opens one of the three doors and reveals that there is nothing behind that door.
- He gives you a choice: switch doors, or stick with the door you chose initially.

Question: Do you switch??

Monty Hall Problem

Let's say we don't switch. Then we are not letting what Wayne told us influence our decision-making. Naively, we know that one of the three doors has a car, hence:

$$P(\text{choosing a car}) = \frac{\text{number of doors with car behind them}}{\text{number of doors}} = \frac{1}{3}$$

Monty Hall Problem

Let's say we do switch. Then the switch is conditional on what Wayne told us. Let D_i be the event the car is behind door i and you picked door 1. Then, we can write the new probability using the law of total probability:

$$\begin{aligned}P(\text{choosing a car}) &= P(\text{get car}|D_1)P(D_1) + P(\text{get car}|D_2)P(D_2) + P(\text{get car}|D_3)P(D_3) \\&= 0 \times \frac{1}{3} + 1 \times \frac{1}{3} + 1 \times \frac{1}{3} \\&= \frac{2}{3}\end{aligned}$$

Suppose we chose door 1 initially. Then, if we do switch and the car is behind door 1, our probability is 0. However, if Wayne reveals either door 2 or door 3 and the car exists in the other door, switching guarantees that we will choose the door with the car inside.

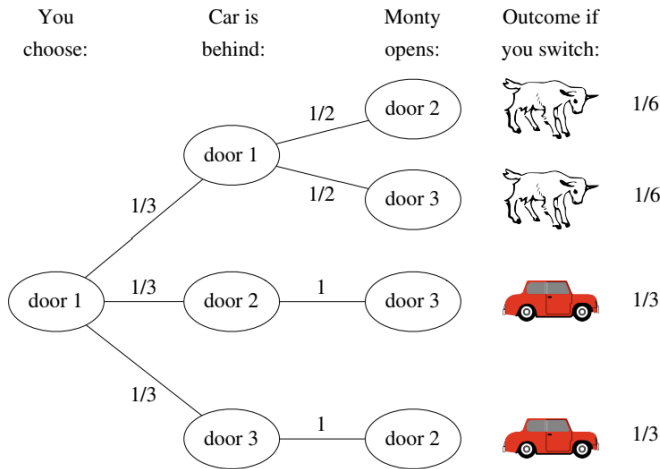
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Another way to think about this: one door has already been revealed to us as junk and we have two doors left. If we do switch, two of those doors are accounted for as a success. If we don't switch, we are under the original notion that just one of the three has a car behind it.

Monty Hall Problem



Independence

Suppose two events A and B have no impact on another. We say that these events are **independent** mathematically if:

$$P(A \cap B) = P(A) \times P(B)$$

otherwise, we call the two events dependent.

Examples:

- First coin toss vs. second coin toss (same events in succession)
- Doing well on a test vs. a tree getting knocked down in the city over (no association)

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Note: This is different from disjoint events. Having disjoint events means that both events cannot happen at the same time (e.g. $A \cap B = \emptyset$).

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Note: For independent events, the occurrence of one event happening does not change the chance of another event occurring.

Activity: Conditional Probabilities

Take 1-2 minutes to rewrite the probabilities of the intersection of events A and B knowing that these two events are independent. What do you get as a result?

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

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Results: Conditional Probabilities

Applying independence, we get the following simplified expressions:

$$P(B|A) = P(B)$$

$$P(A|B) = P(A)$$

Example: Coin Flips

Suppose I flip two coins, one after another. Let A be the event I get a heads on the first flip and B be the event I get a tails on the second flip. What is $P(A \cap B)$?

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$$\begin{aligned} P(A \cap B) &= P(A) \times P(B) \\ &= \frac{1}{2} \times \frac{1}{2} \\ &= \frac{1}{4} \end{aligned}$$

Example: Coin Flips

Suppose I flip two coins, one after another. Let A be the event I get a heads on the first flip and C **be the event I get the same result on the second flip**. What is $P(A \cap C)$?

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Suppose I flip two coins, one after another. Let A be the event I get a heads on the first flip and C **be the event I get the same result on the second flip**. What is $P(A \cap C)$?

As you see, event C is dependent on event A . So, we use conditional probability to help us out here:

$$\begin{aligned} P(A \cap C) &= P(C|A) \times P(A) \\ &= \frac{1}{2} \times \frac{1}{2} \\ &= \frac{1}{4} \end{aligned}$$

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Independence (or not) will determine how you solve a particular probability.

Proposition 2.5.3

For events A and B : if they are independent, then the following events are also independent:

- A and B^C
- A^C and B
- A^C and B^C

Outside of class: Show that each of pair of events are independent using multiplication and compliment rules, as well as the independence definition.

Independence of Three Events

Three events A , B , and C are independent if all statements hold true:

$$P(A \cap B \cap C) = P(A) \times P(B) \times P(C)$$

$$P(A \cap B) = P(A) \times P(B)$$

$$P(A \cap C) = P(A) \times P(C)$$

$$P(B \cap C) = P(B) \times P(C)$$

This means we should also have pairwise independence between all pairwise combinations of events.

- Pairwise independence does not imply independence for 3 events
 - ▶ See example 2.5.5 from *Intro to Probability*
- Can expand to n events, which will lead into our conversation about the Binomial distribution

Activity: Hot Hand

The Hot Hand Fallacy is the idea that an athlete will have streaks of success (or failure) that are higher than their average performance. It's an idea that has been research upon and debated for decades (and perhaps something you've debated when watching a game).

Activity: Hot Hand

Discuss and work on problems in groups of 2-3 related to the fallacy. NBA player Devin Booker has a 33% chance to make a 3-point shot when given the opportunity.

- ① Assume each shot is independent of each other. Let events A , B , and C be the events of him making his first, second, and third shots.
 - ▶ Find $P(A \cap B \cap C)$.
 - ▶ Find $P(A \cap B^C \cap C)$.
 - ▶ Find $P(A \cap B)$ and $P(A \cap C)$.
- ② Now, assume that if he make his previous shot, his percentage of making the next shot is 4 percentage points higher than making his previous shot (with a cap of 1). Then,
 - ▶ Find $P(A \cap B \cap C)$.
 - ▶ Find $P(A \cap B^C \cap C)$.
 - ▶ Find $P(A \cap B)$ and $P(A \cap C)$.

Conditional Independence

For some events B_1 and B_2 given some A which has $P(A) > 0$, we say that B_1 and B_2 are conditionally independent if:

$$P((B_1 \cap B_2)|A) = P(B_1|A) \times P(B_2|A)$$

- Conditional independence does not imply independence (and vice versa)
 - ▶ e.g. mom and dad have genetic disorder, and child has genetic disorder
 - ▶ Look through examples 2.5.9-2.5.11 from *Intro to Probability*

Formulas from Ch. 2

Conditional Probability:

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Multiplication Rule:

$$P(A \cap B) = P(B|A) \times P(A)$$

Law of Total Probability:

$$P(B) = \sum_{j=1}^n P(B|A_j)P(A_j)$$

Bayes' Theorem:

$$P(A_j|B) = \frac{P(A_j \cap B)}{P(B)} = \frac{P(B|A_j)P(A_j)}{\sum_{i=1}^n P(B|A_i)P(A_i)}$$

Formulas from Ch. 2

Independence:

$$P(A \cap B) = P(A) \times P(B)$$

Conditional Independence:

$$P((B_1 \cap B_2)|A) = P(B_1|A) \times P(B_2|A)$$

Paradoxes

As seen with the Monty Hall problem, the existence of conditional probability unearths paradoxes, which change our way of thinking about various probabilities. Here are some notable ones:

- Prosecutor's fallacy (via flipping events in a conditional probability)
- Simpson's paradox (trend of distribution of event probabilities given events different from overall trend)
 - ▶ See medical testing problem from earlier

Exam Notes for Next Tuesday

- Visit the Materials webpage on my website for a study guide
- 75 minutes, closed-note, no electronic devices (aside from calculator).
But, you do get access to:
 - ▶ One 8.5x11" sheet of paper of notes, hand-written or typed
- 4 questions! Questions will be similar in length to the ones seen in checkpoints
- 15% of grade (10% if lowest of the collective three)
- Preparation:
 - ▶ Practice, practice, practice!
 - ▶ Office hours today, Friday, and Monday
 - ▶ Review session on Thursday (come with questions!)

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